

7.3.2 Reuse Asset Management Process

NOTE — The Reuse Asset Management Process governs the **full life cycle of reusable assets**, from inception to retirement. It ensures that assets are properly classified, stored, retrieved, maintained, versioned, and retired, supporting efficient and controlled reuse across projects and domains.

7.3.2.1 Purpose

The purpose of the Reuse Asset Management Process is to **manage the life cycle of reusable assets from conception to retirement**, ensuring that assets are properly classified, controlled, maintained, and made available for effective reuse.

7.3.2.2 Outcomes

As a result of successful implementation of the Reuse Asset Management Process:

- a) an asset management strategy is documented;
 - b) an asset classification scheme is established;
 - c) criteria for asset acceptance, certification, and retirement are defined;
 - d) an asset storage and retrieval mechanism is operated;
 - e) the use of assets is recorded;
 - f) changes to the assets are controlled; and
 - g) users of assets are notified of problems detected, modifications made, new versions created, and deletions of assets from the storage and retrieval mechanism.
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7.3.2.3 Activities and Tasks

The organization shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Reuse Asset Management Process.

7.3.2.3.1 Process Implementation

This activity consists of the following tasks:

7.3.2.3.1.1 Asset Management Planning

The **Asset Manager** shall create an **Asset Management Plan** to define the resources, responsibilities, and procedures for managing assets throughout their life cycle. The plan shall include:

- roles and responsibilities (e.g., asset manager, domain engineer, reuse program administrator);
 - procedures for asset acceptance, classification, storage, retrieval, modification, notification, and retirement;
 - tools and repositories used to manage reusable assets;
 - review and audit mechanisms.
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7.3.2.3.1.2 Asset Management Plan Execution

The asset manager shall execute the plan, ensuring consistent application of defined procedures across all managed assets.

7.3.2.3.1.3 Plan Review

The asset management plan shall be reviewed in accordance with the **Software Review Process (7.2.6)**. Reuse program administrators and domain engineers shall participate in these reviews to ensure alignment with organizational reuse objectives.

7.3.2.3.2 Asset Storage and Retrieval Definition

This activity consists of the following tasks:

7.3.2.3.2.1 Storage and Retrieval Mechanism Implementation

The asset manager shall **implement and maintain an asset storage and retrieval mechanism** (e.g., asset library, repository, registry) that supports secure, organized, and efficient access to reusable assets.

7.3.2.3.2.2 Asset Classification Scheme

The asset manager shall **develop, document, and maintain a classification scheme** for assets. The scheme may include:

- functional domain;
- maturity level (e.g., prototype, certified, retired);

- quality or certification status;
 - applicable standards;
 - intended usage context.
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7.3.2.3.2.3 Storage and Retrieval Reviews

The asset manager shall conduct periodic reviews of the storage and retrieval mechanism in accordance with the **Software Review Process (7.2.6)**. Reuse program administrators and domain engineers shall be included in these reviews to ensure relevance, usability, and completeness.

7.3.2.3.3 Asset Management and Control

For each asset, this activity consists of the following tasks:

7.3.2.3.3.1 Asset Acceptance and Certification

Each submitted asset shall be **evaluated against asset acceptance and certification criteria**, which may include quality, documentation completeness, licensing, and compliance with reuse standards.

7.3.2.3.3.2 Asset Availability

For each accepted asset, it shall be **made available for reuse** through the asset storage and retrieval mechanism in accordance with established access and security policies.

7.3.2.3.3.3 Asset Classification

Each accepted asset shall be **classified** according to the established reuse classification scheme to support discoverability and appropriate reuse decisions.

7.3.2.3.3.4 Configuration Management of Assets

The asset manager shall perform **configuration management** of each asset using the **Software Configuration Management Process (7.2.2)**. This includes version control, change tracking, and baselining.

7.3.2.3.3.5 Reuse Tracking

The asset manager shall **track each reuse** of the asset and report relevant information (e.g., projects, versions used, usage contexts) to the **Domain Engineer** and reuse program administrators.

7.3.2.3.3.6 Notification of Asset Changes

The asset manager shall **monitor and record asset-related requests and problem reports**.

All **asset reusers and the domain engineer shall be notified** of:

- problems detected in the asset;
 - modifications made;
 - new versions created;
 - deletions of the asset from the storage and retrieval mechanism.
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7.3.2.3.3.7 Asset Retirement

Assets shall be **retired** from the storage and retrieval mechanism according to defined retirement procedures and criteria. Retirement may be triggered by obsolescence, quality degradation, security vulnerabilities, or replacement by newer assets.

7.3.2.4 Deliverables

The following deliverables shall be produced during this process:

- Asset Management Plan;
 - Asset classification scheme and taxonomy;
 - Storage and retrieval mechanism implementation and configuration documentation;
 - Asset acceptance and certification records;
 - Asset version histories and configuration management records;
 - Reuse usage logs and reports;
 - Notifications of asset changes and retirements;
 - Asset retirement records.
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7.3.2.5 Audit and Configuration Control

All reusable assets, their metadata, and associated management artifacts shall be placed under appropriate **configuration management**.

Audit records of asset acceptance, modifications, reuse, and retirement shall be maintained according to organizational and regulatory requirements.

Access to asset repositories shall be logged and controlled. Periodic audits shall ensure integrity, consistency, and completeness of stored assets.

Cross-References

This process interacts with the following:

- **Domain Engineering Process (7.3.1)** — provides domains, models, and reusable asset definitions.
- **Reuse Program Management Process (7.3.3)** — defines organizational strategy and policies for reuse.
- **Software Configuration Management (7.2.2)** — provides version control and change management for assets.
- **Software Review Process (7.2.6)** — used for plan and mechanism reviews.
- **Software Problem Resolution (7.2.8)** — may be invoked to handle problems with reusable assets.
- **Software Verification (7.2.4)** and **Validation (7.2.5)** — may provide evidence for asset certification.